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(54) **MICROPILLAR-ENABLED THERMAL
GROUND PLANE**

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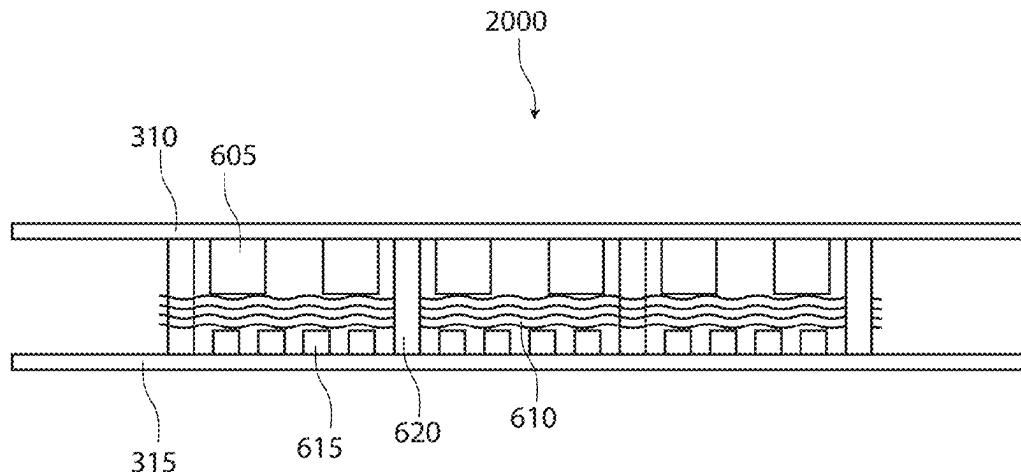
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ABSTRACT

A thermal ground plane (TGP) is disclosed. A TGP may include a first planar substrate member comprising copper and a second planar substrate member comprising a metal, wherein the first planar substrate member and the second planar substrate member enclose a working fluid. The TGP may include a first plurality of pillars disposed on an interior surface of the first planar substrate and a mesh layer disposed on the top of the first plurality of pillars, wherein the mesh layer comprises at least one of copper, polymer encapsulated with copper, or stainless steel encapsulated with copper. The TGP may also include a second plurality of pillars disposed on an interior surface of the second planar substrate member within an area defined by the perimeter of the second planar substrate member and the second plurality of pillars extend from the second planar substrate member to the mesh layer.

21 Claims, 20 Drawing Sheets

(11 of 20 Drawing Sheet(s) Filed in Color)



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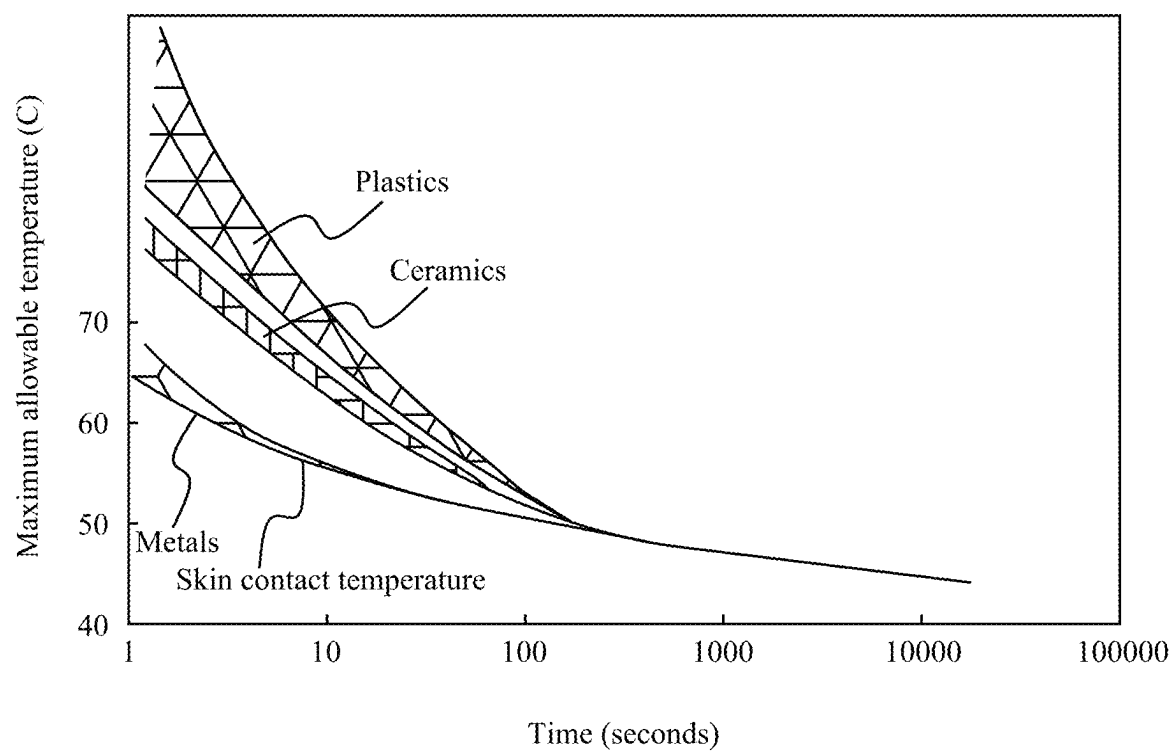
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*Figure 1*

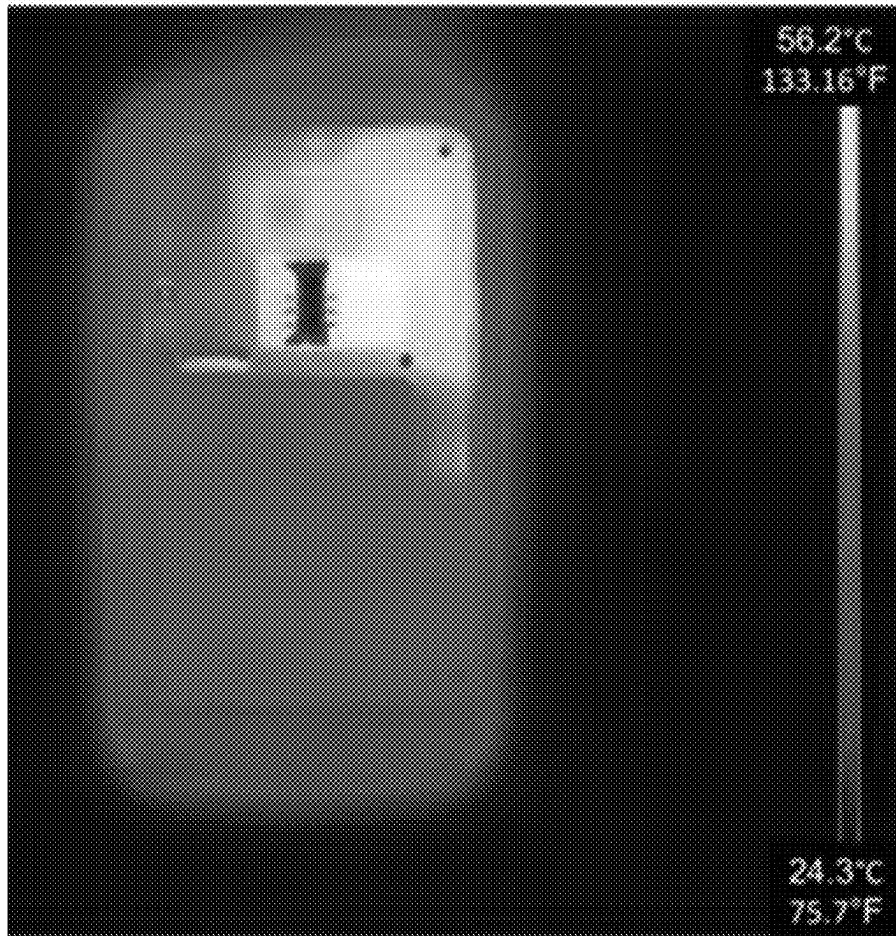


Figure 2

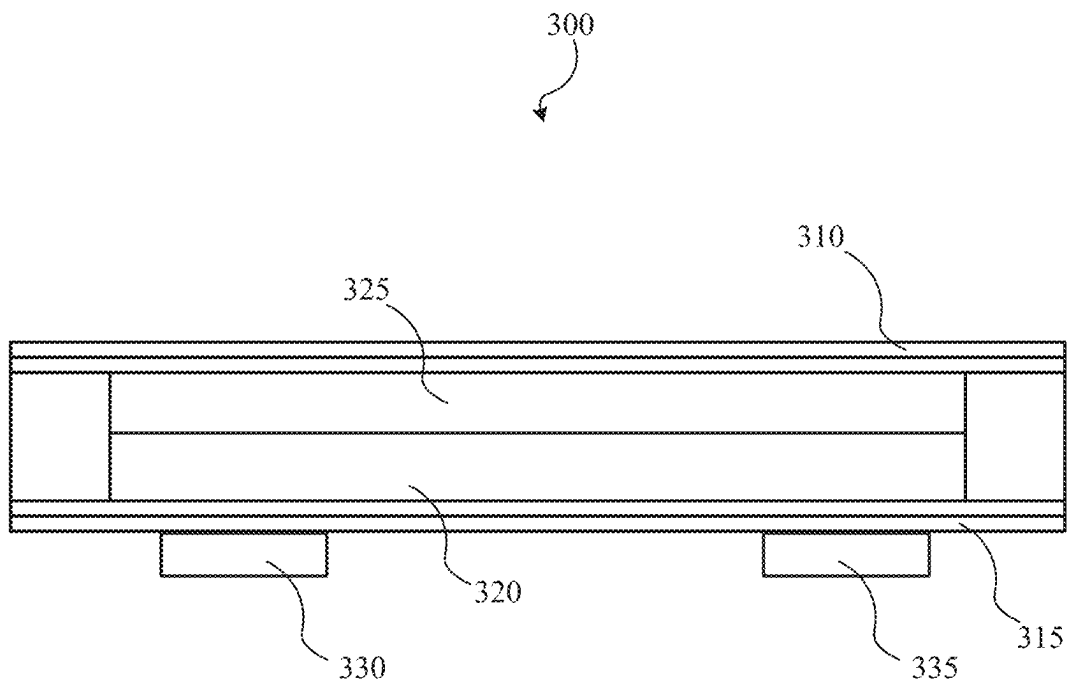


Figure 3

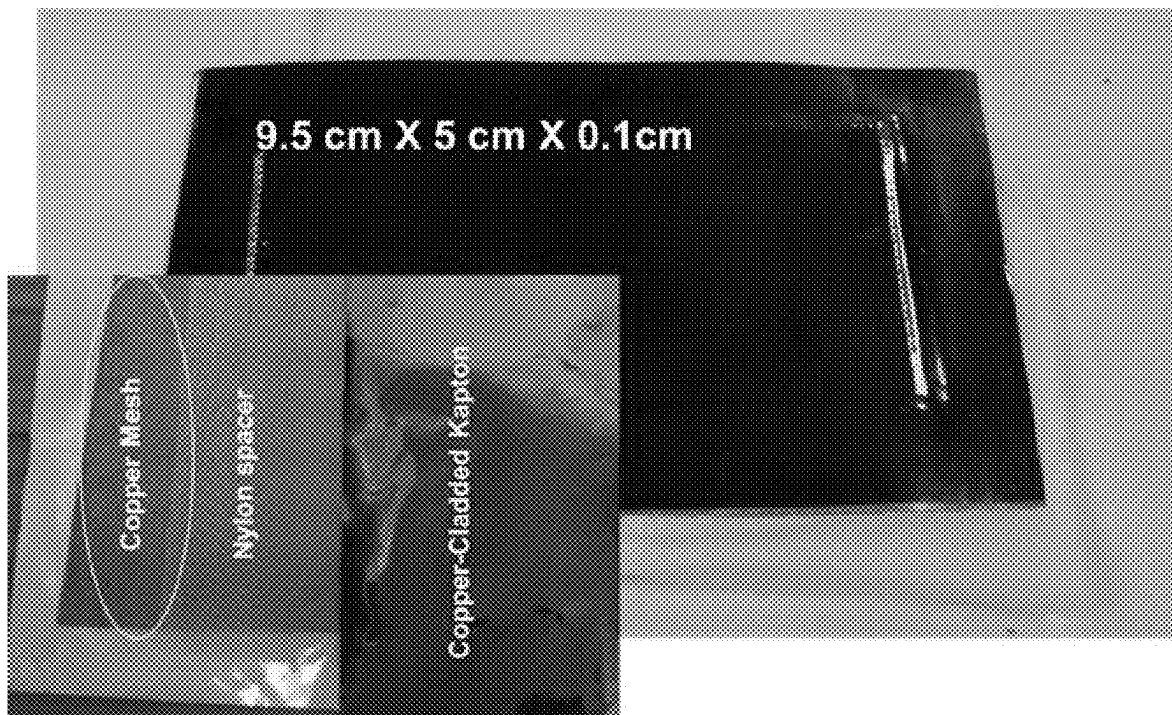
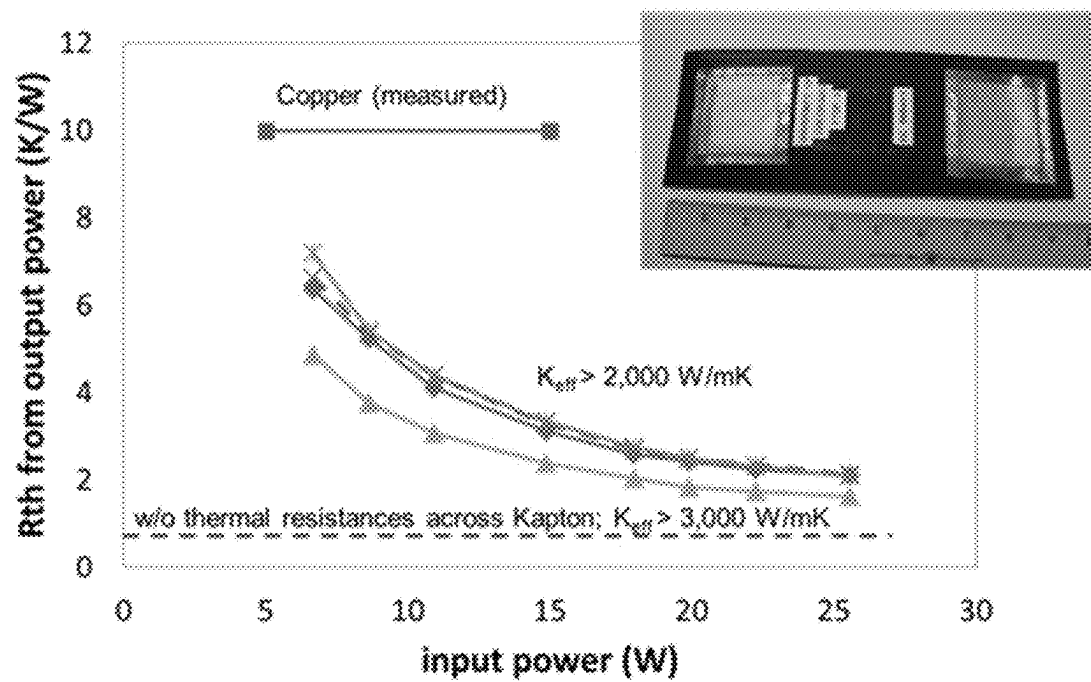


Figure 4

*Figure 5*

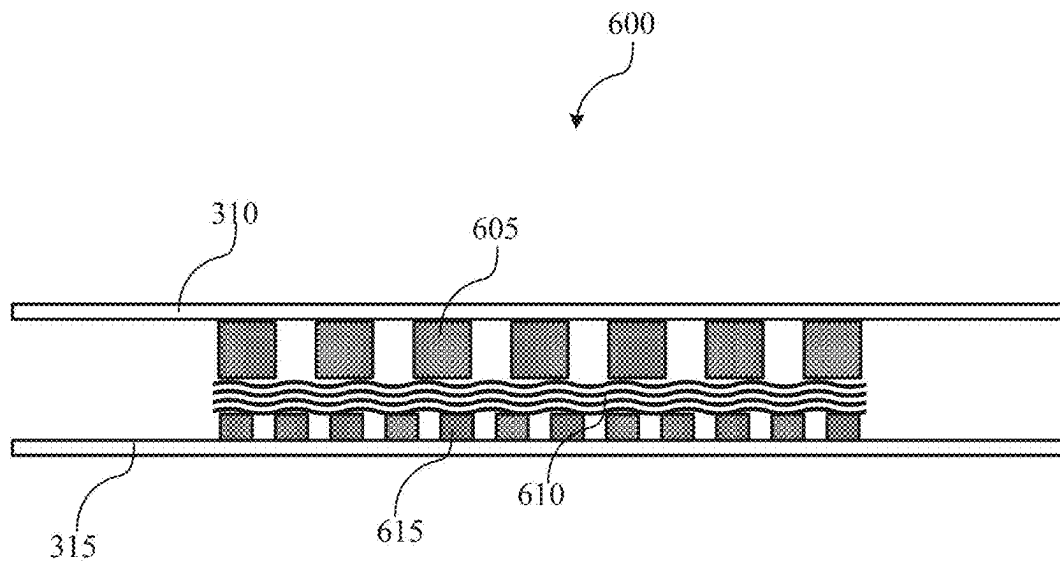


Figure 6

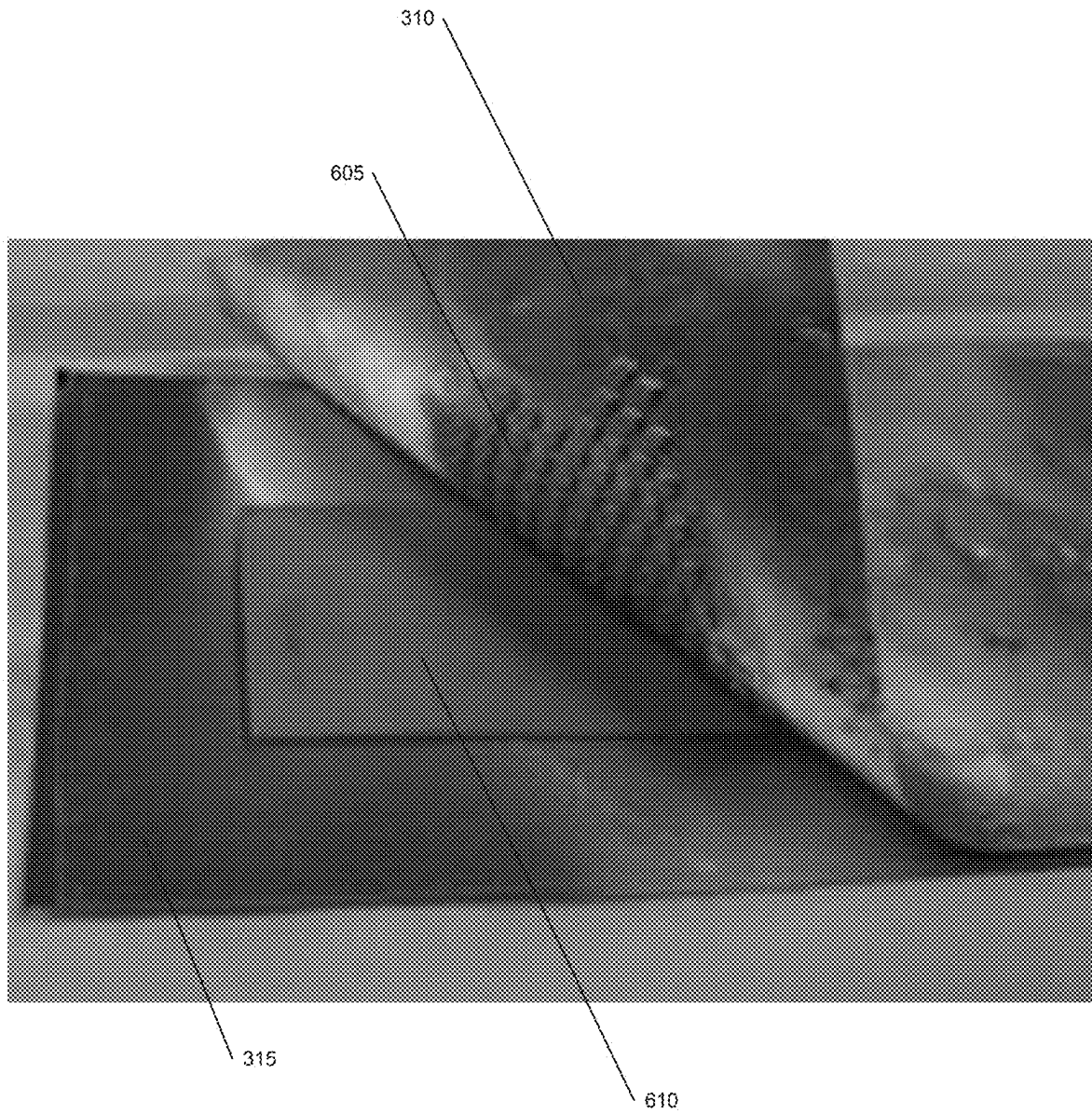


Figure 7

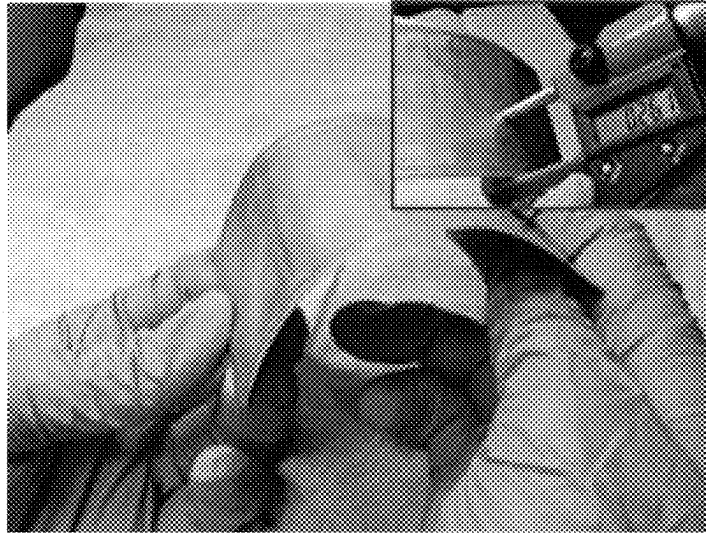
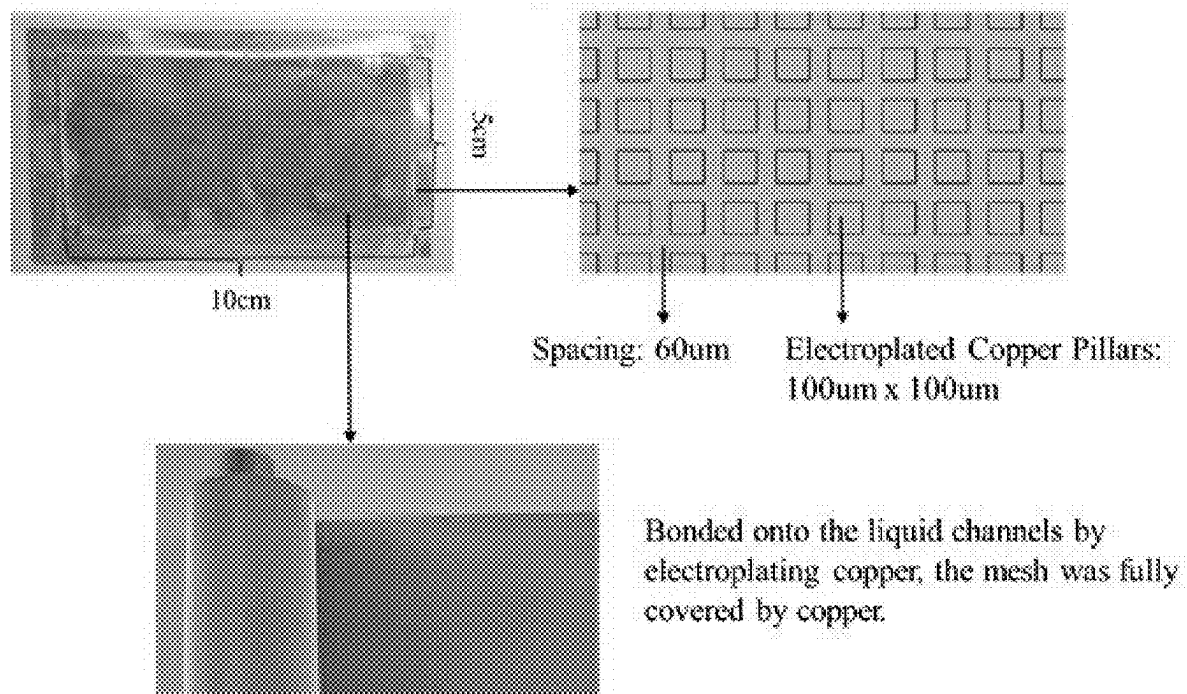


Figure 8

*Figure 9*

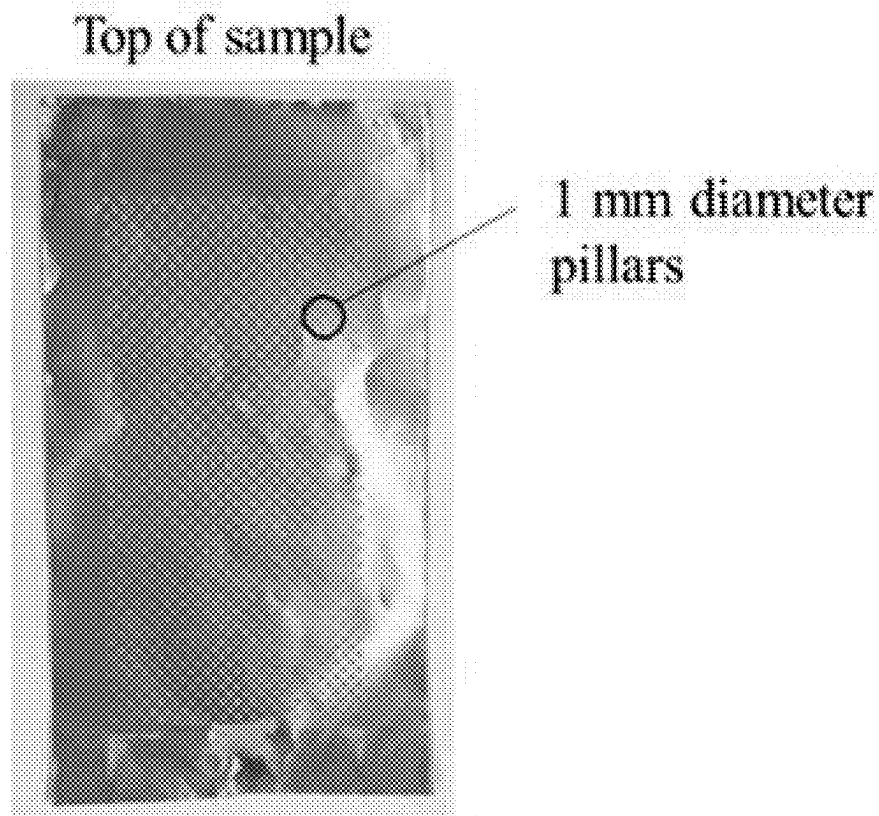


Figure 10

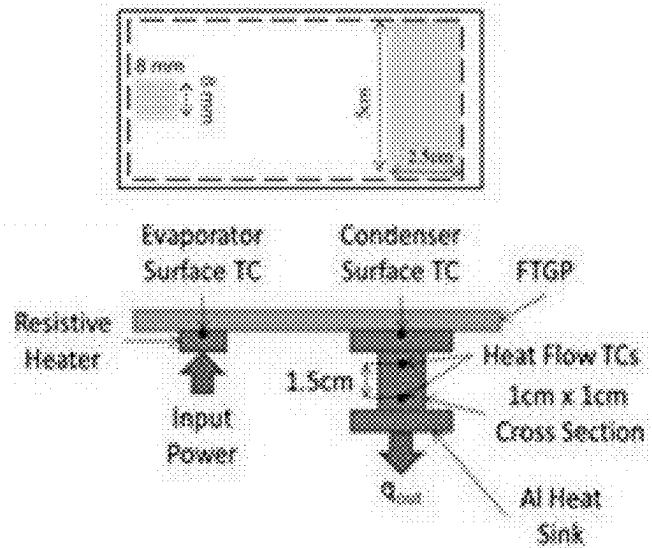
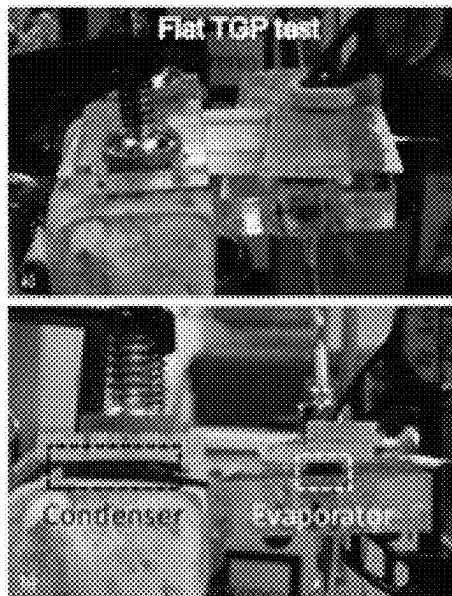
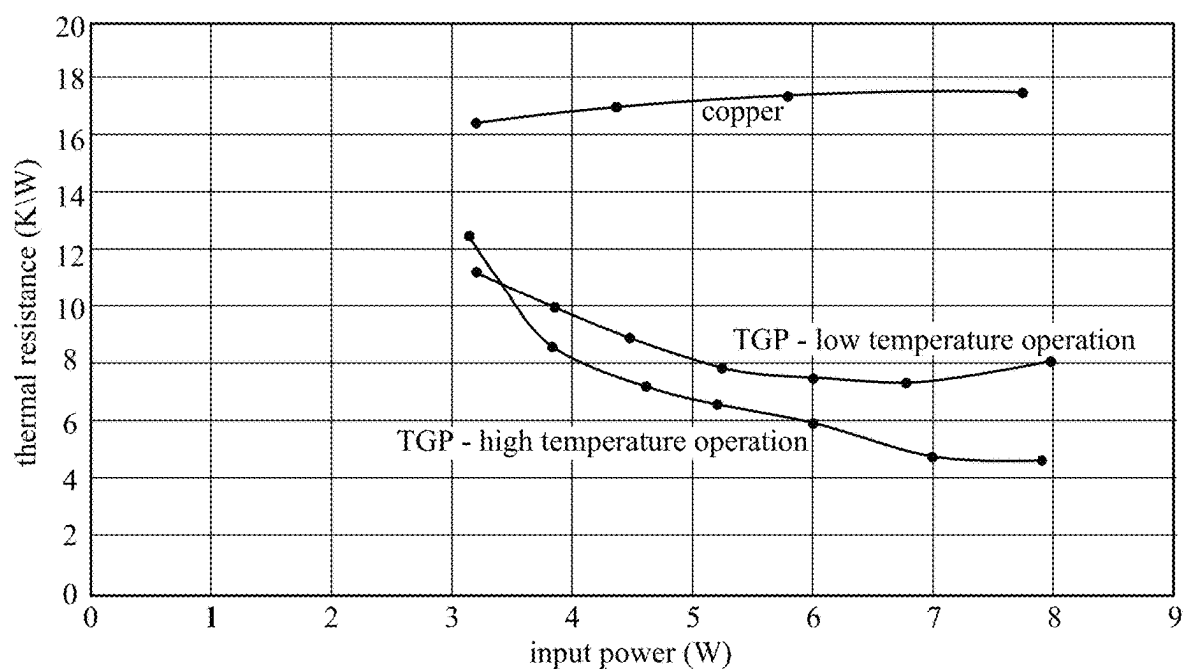
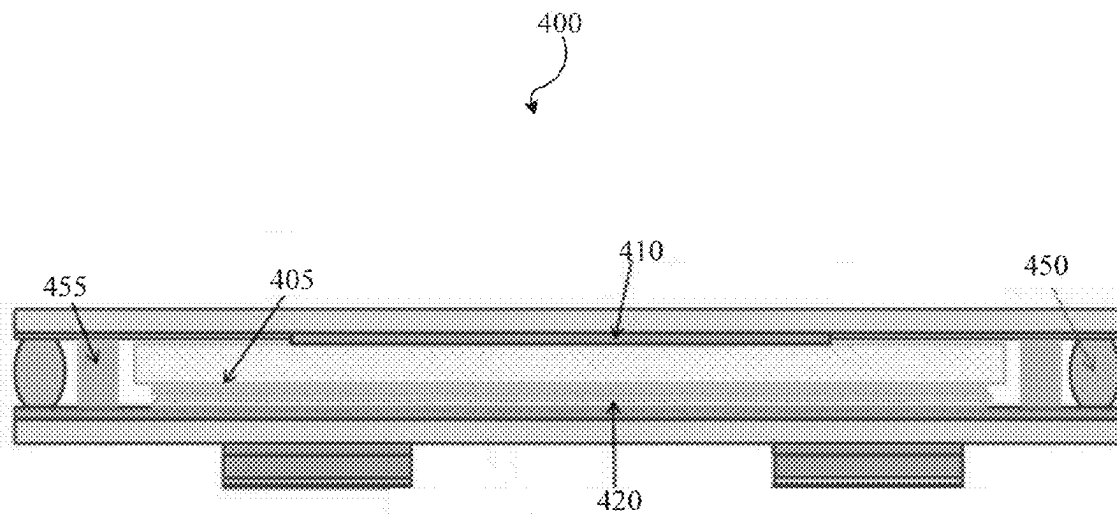
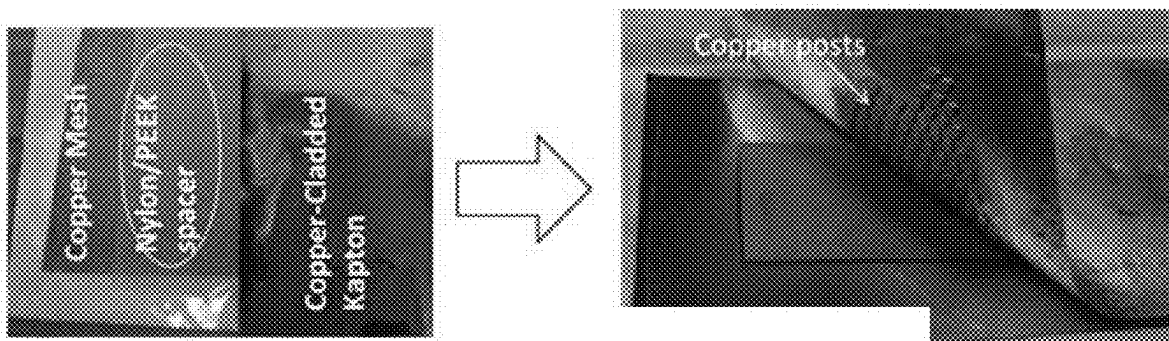
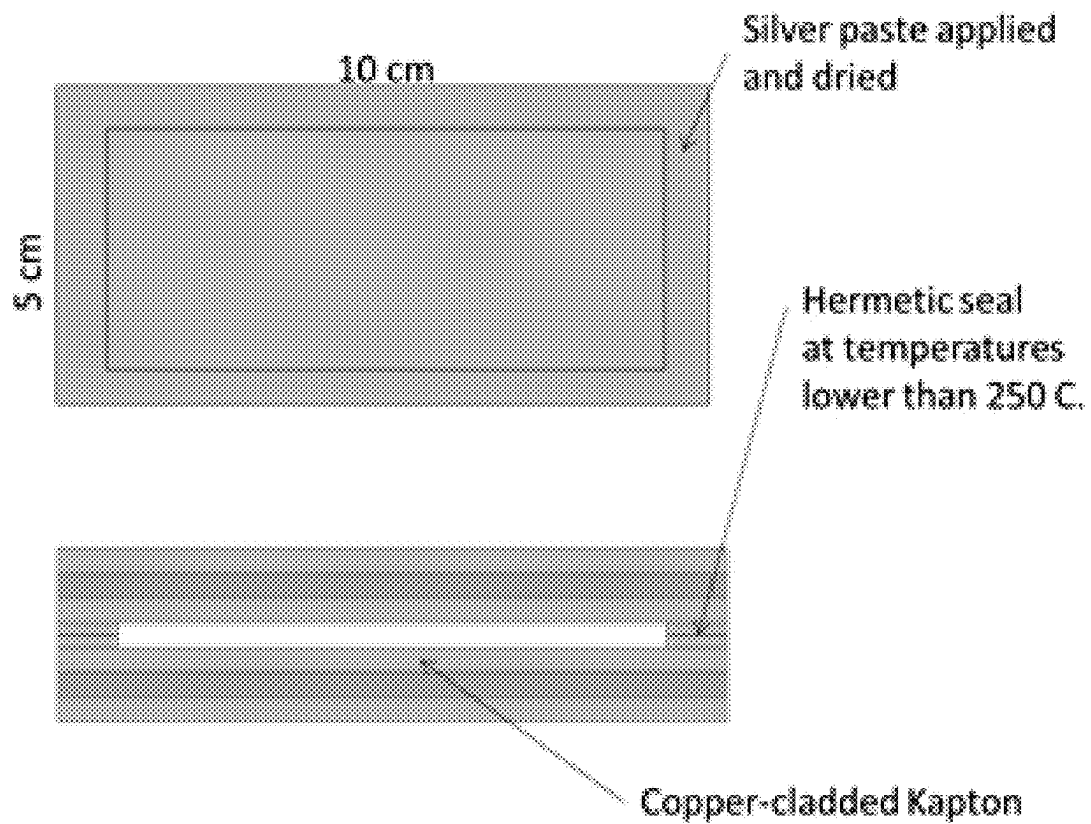


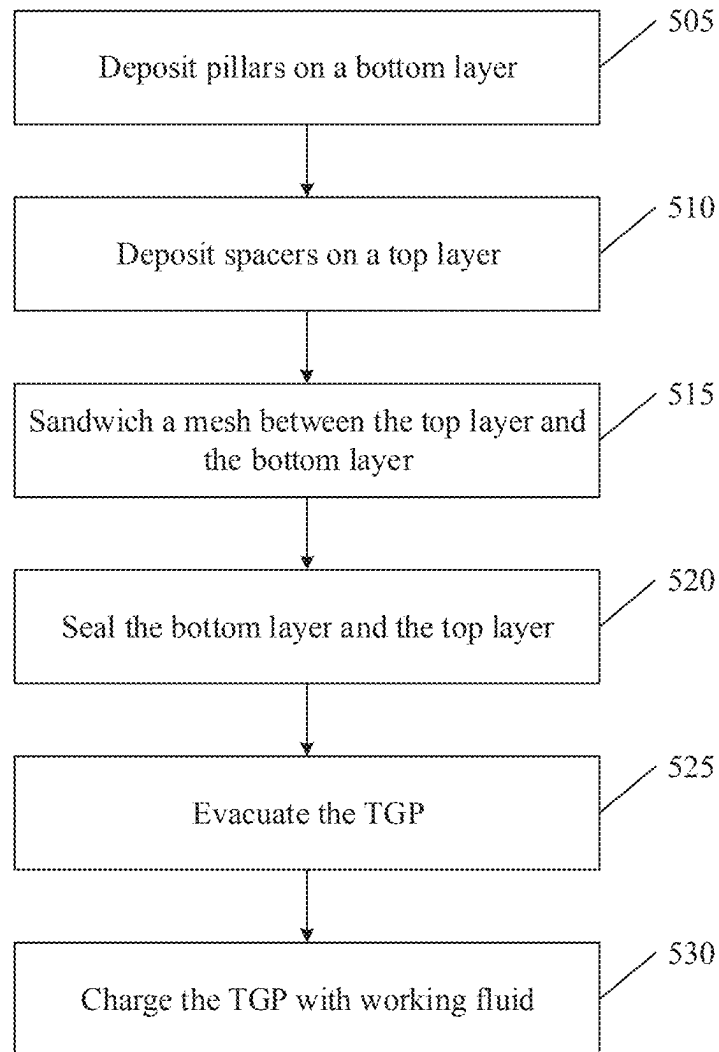
Figure 11

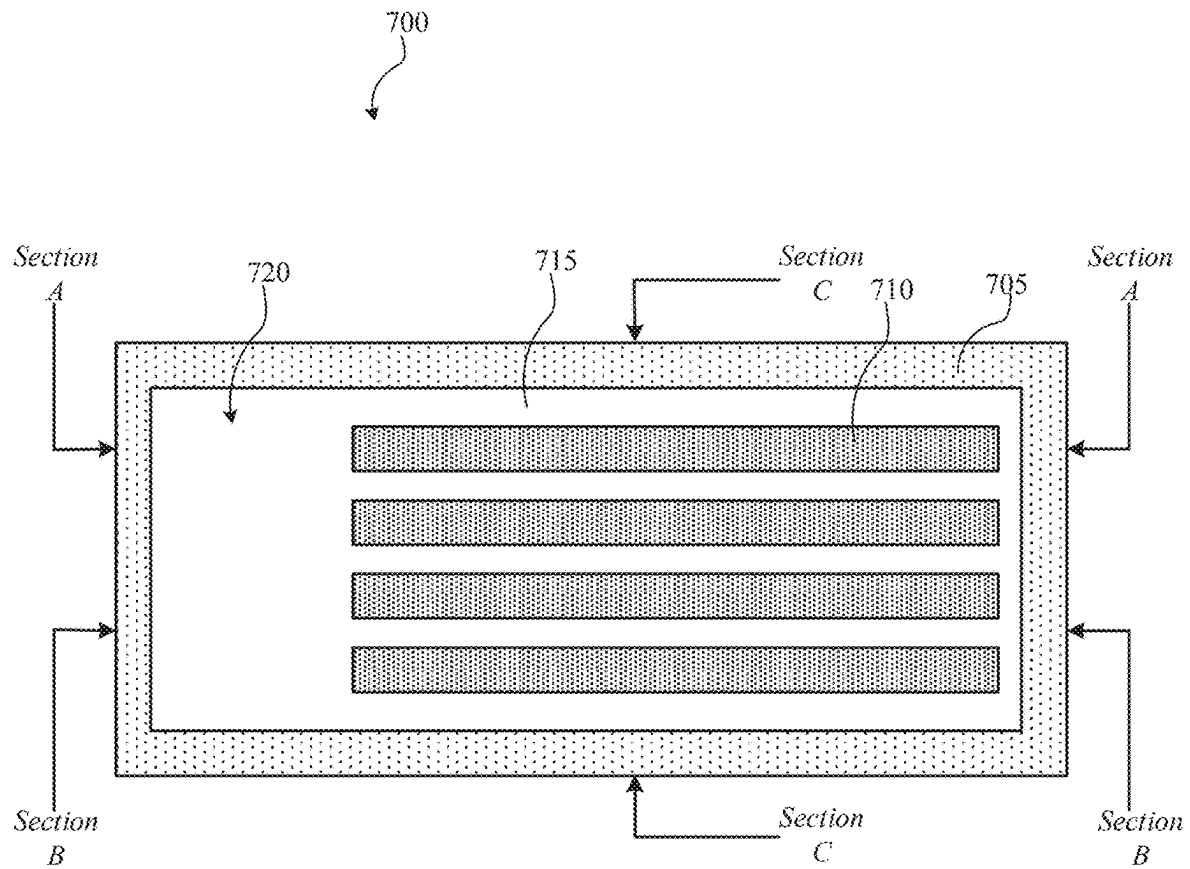
*Figure 12*

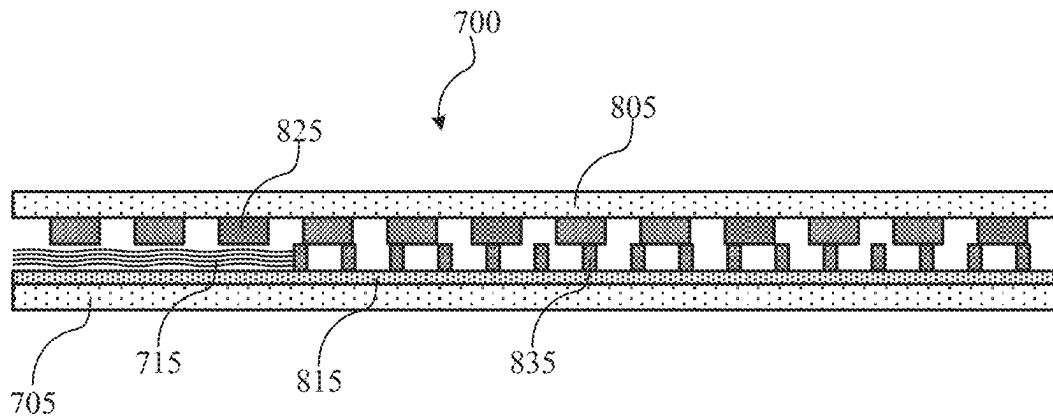
*Figure 13*

*Figure 14A**Figure 14B*

*Figure 15*

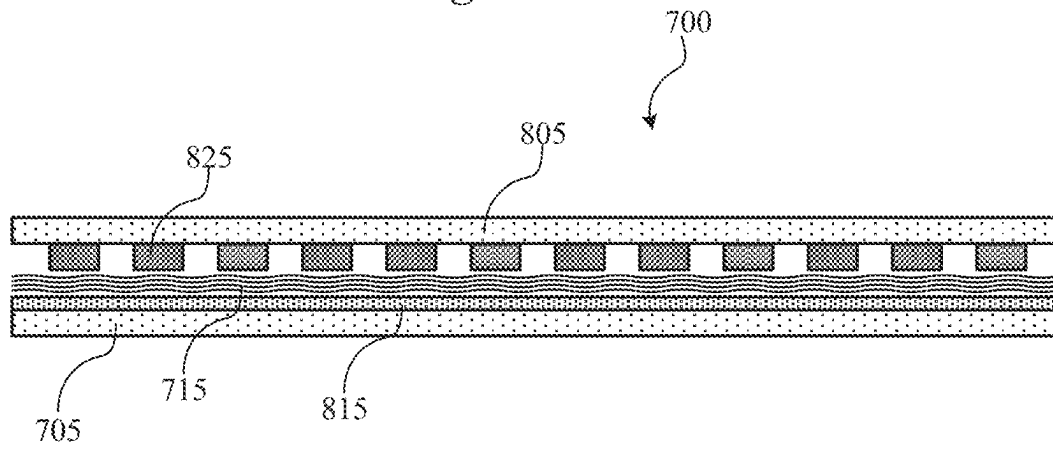
*Figure 16*

*Figure 17*



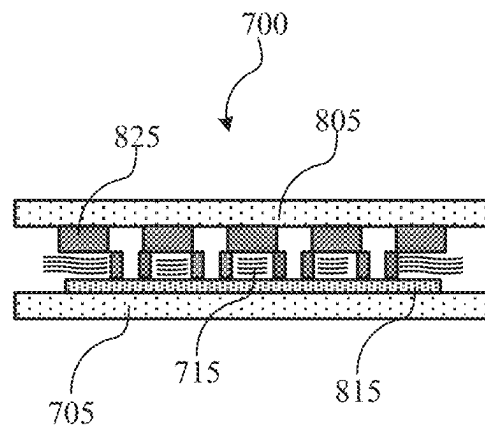
(Section A)

Figure 18A



(Section B)

Figure 18B



(Section C)

Figure 18C

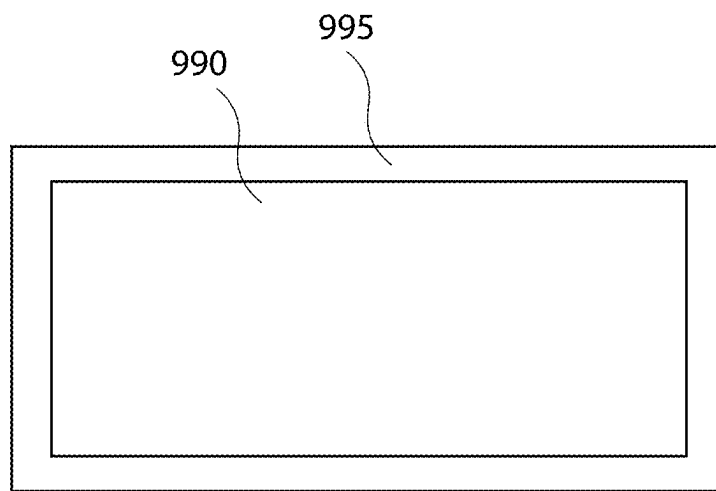


Figure 19

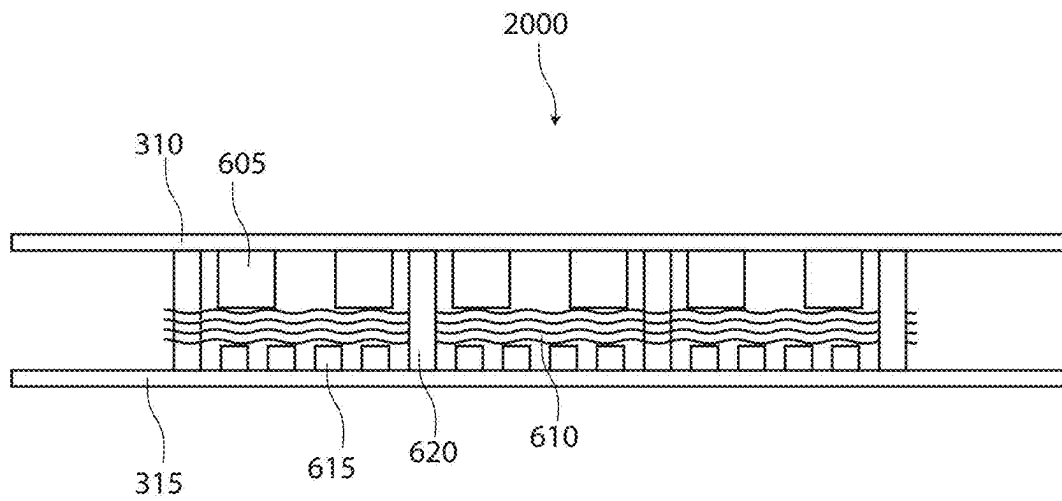


Figure 20

MICROPILLAR-ENABLED THERMAL GROUND PLANE

CROSS-REFERENCE TO RELATED APPLICATIONS

This application is a continuation in part of patent application Ser. No. 14/857,567, filed Sep. 17, 2015, titled MICROPILLAR-ENABLED THERMAL GROUND PLANE, which claims priority from U.S. Provisional Patent Application No. 62/051,761, filed Sep. 17, 2014, titled MICROPILLAR-ENABLED THERMAL GROUND PLANE; and a non-provisional of U.S. Provisional Patent Application No. 62/069,564, filed Oct. 28, 2014, titled POLYMER-BASED MICROFABRICATED THERMAL GROUND PLANE, each of which are incorporated into this disclosure by reference in their entireties.

SUMMARY

A thermal ground plane (TGP) is disclosed. The TGP may include a first planar substrate member configured to enclose a working fluid; a second planar substrate member configured to enclose the working fluid; a plurality of wicking structures disposed on the first planar substrate; and one or more planar spacers disposed on the second planar substrate. The first planar substrate and the second planar substrate may be hermetically sealed.

In some embodiments, the plurality of wicking structures comprise a plurality of pillars or channels disposed on the first planar substrate member and/or the plurality of wicking structures comprise a plurality of pillars or channels formed on the second planar substrate member.

In some embodiments, the first planar substrate member and/or the second planar substrate member comprise a metallic layer, a copper-cladded polymer layer with thermal vias, a copper-cladded polymer layer without thermal vias, a polymer-coated copper layer, a metallic layer encapsulated by a hydrophilic coating, a copper metallic layer encapsulated by a hydrophilic hydrophobic coating, and/or a polymer layer encapsulated by a metallic layer, a polymer layer encapsulated by a hydrophilic coating, and/or a polymer layer encapsulated by a hydrophobic coating. In some embodiments, the first planar substrate member (e.g., a metal cladding layer) and/or the second planar substrate member may be thicker in one or more regions such as, for example, regions near the evaporator or the condenser.

In some embodiments, the plurality of wicking structures comprise a plurality of copper pillars bonded with a mesh layer.

In some embodiments, the plurality of wicking structures may comprise a material with at least one property selected from the list consisting of copper mesh, stainless steel mesh, metal mesh, polymer mesh, copper-encapsulated mesh, a mesh encapsulated by a hydrophilic coating, a mesh encapsulated by a hydrophobic coating, and a mesh encapsulated by a hermetic seal.

In some embodiments, the plurality of wicking structures comprise copper pillars bonded with a stainless steel mesh encapsulated by copper or a copper mesh.

In some embodiments, the one or more planar spacers comprise copper pillars bonded with a stainless steel mesh encapsulated by copper or a copper mesh.

In some embodiments, the one or more planar spacers comprise a plurality of copper pillars or channels.

In some embodiments, the one or more planar spacers comprise a plurality of pillars or channels with various star-shaped cross sections, such as rectangular, circular, and/or star-shaped.

In some embodiments, the plurality of wicking structures comprise polymer pillars hermetically sealed by copper or other coatings, and/or enhanced with a hydrophilic coating or a hydrophobic coating.

In some embodiments, the plurality of wicking structures comprise a mesh layer. The mesh layer, for example, may include a mesh selected from the list consisting of copper mesh, stainless steel mesh, metal mesh, polymer mesh and copper-encapsulated mesh. The mesh layer, for example, may include a hydrophilic coating or a hydrophobic coating or hermetic coating.

In some embodiments, the plurality of wicking structures and/or the one or more planar spacers are deposited using a lithographic patterning process.

In some embodiments, the hermetical seal can be an ultrasound-welded, an electrostatic-welded, or a laser-welded copper-to-copper interface.

In some embodiments, the hermetical seal can include a seal selected from the list consisting of a copper-silver sintered interface, a tin/lead solder, and a lead-free solder alloy.

A thermal ground plane (TGP) is disclosed. The TGP may include a top layer; a mesh layer comprising a plurality of arteries formed along a length of the mesh layer; and a bottom layer comprising a plurality of pillars that extend into the plurality of arteries/. The bottom layer and the top layer may be sealed around at least one edge of the top layer and at least one edge of the bottom layer. Each of the plurality of pillars may have at least one dimension that is less than a dimension of at least one of the plurality of arteries.

In some embodiments, the plurality of arteries may be cut from the mesh layer. In some embodiments, the TGP may include a micro-wick layer disposed on the bottom layer. In some embodiments, the TGP may include a plurality of pillars disposed on the top layer.

BRIEF DESCRIPTION OF THE FIGURES

The patent or application file contains at least one drawing executed in color. Copies of this patent or patent application publication with color drawing(s) will be provided by the Office upon request and payment of the necessary fee.

These and other features, aspects, and advantages of the present disclosure are better understood when the following Detailed Description is read with reference to the accompanying drawings. The patent or application file contains at least one drawing executed in color. Copies of this patent or patent application publication with color drawing(s) will be provided by the Office upon request and payment of the necessary fee.

FIG. 1 is a graph illustrating the maximum allowable surface temperatures of a mobile system using different materials with different contact duration.

FIG. 2 is an example infrared image of illustrating a non-uniformly heated mobile device without effective heat spreading according to some embodiments.

FIG. 3 illustrates an example diagram of a thermal ground plane (TGP) according to some embodiments.

FIG. 4 illustrates examples of flexible thermal ground planes according to some embodiments.

FIG. 5 illustrates the thermal resistance from an evaporation region to a condensation region of a TGP according to

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some embodiments calibrated with a 1 inch by 1 inch heater injecting heat to the TGP and 1 inch by 1 inch heat sink extracting heat from the TGP.

FIG. 6 illustrates a TGP according to some embodiments.

FIG. 7 shows a TGP with a top layer having pillars formed therein, a bottom layer having pillars formed therein, and a wicking structure according to some embodiments.

FIG. 8 illustrates the thinness of TGPs manufactured according to some embodiments.

FIG. 9 illustrates an example TGP having a bottom layer with copper pillars electroplated and bonded with a copper-coated stainless steel mesh according to some embodiments.

FIG. 10 shows a top layer with a plurality of copper pillars according to some embodiments.

FIG. 11 shows an experimental setup that can be used to characterize the properties of a thin flexible TGP with a condensation region defined by a cold plate according to some embodiments.

FIG. 12 is a graph showing the thermal resistances of a TGP manufactured according to some embodiments and using the experimental setups shown in FIG. 11.

FIG. 13 shows an experimental setup that can be used to characterize the properties of a thin flexible TGP with distributed condensers condensation where the heat is extracted by nature air convection according to some embodiments.

FIG. 14A illustrates another example TGP according to some embodiments.

FIG. 14B shows a TGP according to some embodiments.

FIG. 15 illustrates another example TGP according to some embodiments.

FIG. 16 illustrates an example process for fabricating a TGP according to some embodiments.

FIG. 17 illustrates a top view of another TGP 700 according to some embodiments.

FIGS. 18A, 18B, and 18C illustrate side views of the TGP shown in FIG. 17 according to some embodiments.

FIG. 19 illustrates a block diagram of a wicking structures according to some embodiments.

FIG. 20 illustrates a TGP according to some embodiments.

DETAILED DESCRIPTION

One challenge for mobile systems, e.g., smartphones, tablets and wearable electronics, is the control of the skin temperatures. The skin temperature is the temperature of an exterior portion of a device (e.g., the case) that is touched by fingers, hands, face, ears, or any other part of a human body. When the temperature of a portion of a device reaches beyond the maximum allowable temperature, a user would consider the temperature of the device to be hot. Of course, this perception of heat is dependent on the materials and the duration of the contact; it also varies from one person to another one due to their difference in thermal physiology. FIG. 1 illustrates a graph of acceptable skin temperatures for a number of different materials considering different touch duration.

As illustrated in FIG. 2, a hot spot or region with a much higher temperature than other locations on a smart phone (or other device) could be generated by an electronic chip such as, for example, a 5-Watt processor or a 1-Watt, small-size wireless amplifier. These hot spots or regions could be removed by effective heat spreading since the temperatures in the area outside these hot spots can be much lower.

Typically, a metal heat spreader, such as, for example, an aluminum or copper heat spreader, may be effective. How-

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ever, the thickness of the metal heat spreader is being reduced in order to meet the demand for the reduction of the total system's thickness. For most mobile systems, a thinner configuration is usually desirable. With a thin metal heat spreader, its thermal resistance increases since it is affected by the thermal conductivity and cross-section area for heat conduction. Alternative approaches can use high thermal conductivity graphite heat spreaders. However, the thermal conductivity of graphite heat spreaders generally decreases with the thickness of the graphite layer. The high thermal conductivity graphite, e.g., thermal conductivity of 1,500 W/mK, is too thin, e.g., 0.017 mm, to be effective to transfer heat over a large area.

Some embodiments include a thin thermal ground plane (TGP) with improved thermal performance. For example, a TGP may use phase-change heat transfer mechanisms with an evaporation-vapor transport-condensation-liquid return path for heat transfer. As another example, a TGP may be a very good heat spreader with effective thermal conductivities higher than that of a copper (see FIG. 5). In some embodiments, a TGP may include components and/or may be manufactured with processes described in U.S. patent application Ser. No. 12/719,775, entitled "Flexible Thermal Ground Plane and Manufacturing the Same," which is incorporated herein in its entirety for all purposes.

Embodiments may also include a top layer including a plurality of spacers (or pillars) sealed with a bottom layer having a plurality of pillars. In some embodiments, the spacers may be deposited on the top layer using a lithographic patterning process. In some embodiments, the pillars may be deposited on the bottom layer using a lithographic patterning process. In some embodiments, the pillars and spacers may be bonded or scribed on the first and/or second planar substrate. In some embodiments, the pillars may have a diameter that is less than or greater than the diameter of the pillars and/or spacers. In some embodiments, the spacers pillars may have a spacing that is less than or greater than the spacing diameter of the pillars. In some embodiments, the pillars in the spacers may have a height, diameter or spacing that is less than or greater than the spacing of the pillars in the wicking structure.

FIG. 3 illustrates a diagram of an example TGP 300 according to some embodiments. In this example, the TGP 300 includes a top layer 310, a bottom layer 315, a liquid channel 320, and/or a vapor core 325. The TGP 300, for example, may operate with evaporation, vapor transport, condensation, and/or liquid return of water or other cooling media for heat transfer between the evaporation region 330 and the condensation region 335. The top layer 310 may include copper, polyimide, polymer-coated copper, copper-cladded Kapton, etc. The top bottom layer 315 may include copper, polyimide, polymer-coated copper, copper-cladded Kapton, etc. In some embodiments, the top layer 310 and the bottom layer 315 of the TGP may be sealed together using solder, laser welding, ultrasonic welding, electrostatic welding, or thermo-pressure compression or a sealant 340.

In some embodiments, the liquid channel 320 may include copper mesh, stainless-steel mesh, or meshes made of other materials but encapsulated with copper. The liquid channel 320, for example, may include one, two, three, four, five, six or more layers of mesh of the same or different materials. In some embodiments, the liquid channel 320 may be bonded and/or sealed using FEP (fluorinated ethylene propylene), electroplating, sintering, and/or other adhesives or sealants with the top layer 310. In some embodiments, in order to enhance hydrophilic properties of the metallic mesh, atomic layer deposition (ALD) of Al_2O_3 , TiO_2 , SiO_2 ,

or other coatings may be used to encapsulate meshes of the liquid channel 320 with desirable functionality.

In some embodiments, the vapor core 325 may include a plurality of pillars and/or channels disposed on the top layer 310. The pillars and/or channels may be deposited using any type of deposition technique such as, for example, vapor deposition, etc.

FIG. 4 illustrates examples of flexible thermal ground planes according to some embodiments. The TGPs may include may an active region of various sizes, for example, the TGP may have at least one dimension (e.g., length, width, radius, diameter, etc.) that is less than 2 cm, 5 cm, 10 cm, 20 cm, 50 cm, 100 cm, 500 cm, 1000 cm, etc. and/or a thickness less than 0.1 mm, 0.25 mm, 0.5 mm, 1 mm, 10 mm 50 mm, 100 mm, etc. In a specific example, the active region may have dimensions of 9.5 cm×5 cm×0.1 cm.

In some embodiments, a TGP may include a copper-cladded Kapton bottom layer and a copper-cladded Kapton top layer with a copper or metallic mesh and/or a polymer spacer sandwiched in between. The metallic mesh may be copper mesh, stainless-steel mesh with or without a copper coating, or meshes made of other materials but encapsulated with copper. The copper mesh layer, for example, may include one, two, three, four, five, six or more layers of mesh. In some embodiments, portions of the TGP (e.g., the copper mesh and/or the polymer spacer) may be bonded and sealed using FEP (fluorinated ethylene propylene), electroplating, sintering, and/or other sealants. In some embodiments, in order to enhance hydrophilic properties of the metallic mesh, atomic layer deposition (ALD) of Al_2O_3 , TiO_2 , SiO_2 , or other coatings may be used to encapsulate the meshes with desirable functionality.

FIG. 5 illustrates the thermal resistance from the evaporator evaporation region to the condensation region of a TGP manufactured according to some embodiments. In this example, the TGP includes is characterized with a 1 inch by 1 inch heater injecting the heat to the TGP and a 1 inch by 1 inch heat sink extracting the heat from the TGP. In this example, the active region of the test sample is about 200 mm×50 mm×1 mm. The results are compared with those obtained for a copper reference sample with the same dimensions.

In some embodiments, the TGP may include a layer having a plurality of pillars manufactured with electroplating processes that produces, for example, large bonding pads with copper-nickel-gold layers. Additionally or alternatively, in some embodiments, an ALD hydrophilic or hydrophobic coating process may be used to change the wettability of the pillars.

As shown in FIG. 5, the thermal resistances of a TGP manufactured according to embodiments are compared with those of a copper reference sample with the same dimensions. The thermal resistance from the evaporator to the condenser is a function of input power of the heater. As shown in the graph, the thermal resistance drops from 7 K/W to about 2 K/W when the power increases from 5 to 25 Watts. The corresponding thermal resistance of the copper sample is about 10 K/W. The effective thermal conductivity of the TGP reaches about 2,000 W/mK at 25 Watts since its resistance is about 1/5 of the copper with the thermal conductivity of 400 W/mK. In another test, our customer ran the power all the way to 35 Watts and reached the effective thermal conductivities up to 4,000 to 7,000 W/mK.

In some embodiments, thermal vias with higher thermal conductivity than the substrate materials may be used. However, thermal vias were not used in this sample which was used to create generate the illustration results shown in

the graph in FIG. 5; therefore, there is a relatively large and undesirable thermal resistance across the 50-um thick Kapton layer due to the low thermal conductivity of Kapton. With the use of thermal vias, the total thermal resistance of the TGP may increase with an effective thermal conductivity over 3,000 W/mK even at the 25-Watt power level.

FIG. 6 illustrates a TGP 600 according to some embodiments. In this example, the top layer 310 and/or the bottom layer 315 may include copper cladded Kapton and/or may be similar to the top layer 310 and the bottom layer 315 described above in conjunction with FIG. 3. In some embodiments, the top layer 310 and the bottom layer 315 of the TGP 600 may be sealed together using solder, laser welding, ultrasonic welding, electrostatic welding, or thermo-pressure compression. In between the top layer 310 and the bottom layer 315 the TGP 600 may include a liquid channel, a wicking structure, and/or a vapor core.

The wicking structure 610, for example, may include a copper mesh, stainless-steel mesh with or without a copper coating, or one or more mesh made of other materials but encapsulated with copper. The wicking structure 610, for example, may include one, two, three, four, five, six or more layers of mesh of the same or different material. In some embodiments, the wicking structure 610 may be bonded and/or sealed using FEP (fluorinated ethylene propylene) electroplating, sintering, and/or other adhesives or sealants with the top layer 310. In some embodiments, in order to enhance hydrophilic properties of the wicking structure 610, atomic layer deposition (ALD) of Al_2O_3 , TiO_2 , SiO_2 , or other coatings may be used to encapsulate meshes of the liquid channel 615 with desirable functionality. In some embodiments, the wicking structure 610 may be bonded with or on a plurality of pillars of the liquid channel 615.

In some embodiments, the wicking structure 610 may include a woven mesh that may be bonded to the pillars of the liquid channel 615 and/or the vapor core 605 through an electroplating process. In some embodiments, the mesh may be a copper woven mesh or a metallic mesh. During this process, for example, the wicking structure 610 (e.g., woven mesh) may be encapsulated by copper. The top layer 310 with the vapor core 605 and the bottom layer 315 with the liquid channel 615 may be sealed together with the wicking structure 610 with a mesh in between. After sealing, the TGP 600 may be charged with a working fluid, such as water, methane, ammonia, or other coolants or refrigerants that are compatible with the surfaces exposed to the working fluid.

The liquid channel 615, for example, may include a plurality of pillars (e.g., electroplated pillars).

The vapor core 605, for example, may include a plurality of pillars (e.g., electroplated pillars). In some embodiments, the pillars in the vapor core 605 may have at least one dimension (e.g., height, width, length, diameter, etc.) that is smaller than at least one dimension (e.g., height, width, length, diameter, etc.) of the pillars of the liquid channel 615.

In some embodiments, a photo-lithographic patterning process may be used to fabricate the pillars in the liquid channel 615 and/or the vapor core 605. For example, a photo-lithography process may be used to form a plurality of pillars on the top layer 310 to form the vapor core 605 and/or a photo-lithographic patterning process may be used to form a plurality of pillars on the bottom layer 315 to form the liquid channel 615. The photo-lithographic patterning process, for example, may control the positioning and/or the heights of the pillars with a resolution of a few microns. The plurality of pillars on the top layer 310 and/or on the bottom layer 315 may be formed by the mechanical scribing process.

In some embodiments, the thickness of a TGP (e.g., TGP 300 or TGP 600, or any other TGP described in this disclosure) may be a few millimeters or even less than about 500, 450, 400, 350, 300, 250, 200, 150, or 100 microns. For example, the top layer 310 and/or the liquid channel 615 may be 5, 10, 20, 30, 40, 50, 60, etc. microns thick, the wicking structure (e.g., wicking structure 610) may be 5, 10, 20, 30, 40, 50, 60, etc. microns thick, and/or the vapor core (e.g., vapor core 605) may be 10, 20, 30, 40, 50, 75, 100, 125, 150, etc. microns thick.

FIG. 7 shows an example of a TGP 600 with a top layer 310 having pillars formed thereon to form the vapor core 605, a bottom layer 315 having pillars formed thereon to form the liquid channel 615 and a wicking structure 610 disposed in between according to some embodiments.

In some embodiments, the pillars may be fabricated using electroplating over photo-lithography-defined openings. In some embodiments, TGPs manufactured according to some embodiments may have a thickness larger than or less than 0.25 mm as shown in FIG. 8. The total thickness can be reduced further since all dimensions of critical features are defined by the photo-lithography processes. In some embodiments, TGPs may have a thickness less than about 0.25, 0.2, 0.15, 0.1, 0.05 mm, etc.

In some embodiments, the pillars of the liquid channel 615 and/or the vapor core 605 may allow the TGP to remain effective under different mechanical loadings even in an extremely thin configuration.

FIG. 9 illustrates an example TGP with a plurality of electroplated pillars and mesh according to some embodiments. For example, the pillars on the top layer and/or the bottom layer may be fabricated using a photo-lithographic patterning process and/or may have any dimension such as, for example, 100 μm $\times$ 100 μm squares with 60 μm spacing in between each square. These pillars, for example, may be bonded to a stainless steel woven mesh (e.g., having 500 wires/inch, 50 μm thick) through various bonding techniques such as, for example, copper electroplating. The mesh, for example, may also be at least partially or completely encapsulated by electroplated copper.

In some embodiments, the combined mesh-pillar structure may achieve low capillary radius (or high pumping pressure) in the evaporation regions and higher flow hydraulic radius (or low flow pressure drop) in the fluid channel.

In some embodiments, a controlled over-plating process may be used to form the pillars with rounded heads. In some embodiments, this shape may be beneficial, for example, to form a sharp angle of each interface between a pillar and the mesh bonded. These sharp angles, for example, may enhance the capillary pumping force pulling the liquid returned from the condenser to the evaporator.

FIG. 10 shows an example top layer with copper pillars electroplated through a photolithography-defined opening according to some embodiments. In this example, the height of the pillars is 100 microns, the diameter (or width) of the pillars is 1 mm, and the spacing between the pillars is 2 mm. In this example, the pillars may define the gap of the vapor core. With large spacing (e.g., a spacing between pillars that is greater than the diameter of the individual pillars or a spacing between pillars that is greater than or equal to twice the diameter of the individual pillars) the flow resistance may be reasonably low.

In some embodiments, the pillars formed on the top layer and/or on the bottom layer may be hydrophilic. In some embodiments, the pillars may have a cross section that is round; oval; polygonal; star shaped as shown, for example, in FIG. 6; hexagonal; octagonal; pentagonal; square; rect-

angular; triangular; etc. In some embodiments, condensation of vapor can occur along the vapor core. In some embodiments, the hydrophilic property of the pillars in the vapor core can reduce the size of vapor droplets or bubbles. In some embodiments, the star-shaped pillars can further enhance the hydrophilic properties or wettability.

In some embodiments, the pillars may be constructed from material other than copper. The pillars, for example, may be manufactured from polymer using photo-lithographic deposition and etching techniques. Such polymer pillars, for example, can be baked, followed by hermetic encapsulation with copper, ALD moisture barrier coating or other sealing materials. In some embodiments, the thermal conductivities of polymer pillars may be much lower than that of copper.

In some embodiments, the process of manufacturing a TGP may include a removing air out of the TGP followed by charging the TGP with water or other working fluid that is compatible with an exposed TGP inner surface. In some embodiments, a small diameter copper tube may be coupled with the TGP to allow for vacuuming and/or charging. After water charging, for example, this tube can be pinch-sealed. In some embodiments, the pinch-sealed tube can be removed by an additional seal separating the active region from the region with the tube.

FIG. 11 shows an experimental setup that can be used to characterize the properties of a thin flexible TGP according to some embodiments. In this example, the evaporator (e.g., heater) region was defined by the heater size was of 8 mm $\times$ 8 mm and the condensation region was defined by the condenser block was with an area of 5 cm $\times$ 2.5 cm. The temperature difference between the evaporator evaporation region and the condenser condensation region was measured, and this difference per unit of heat transferred to the condenser heater's input power was used to calculate considered the thermal resistance. The entire setup was covered by thermal insulation to reduce air natural convection effect by air.

FIG. 12 is a graph showing the thermal resistances as a function of heater's input power of a TGP manufactured according to some embodiments and using the experimental setups shown in FIG. 11. As shown in the figures, TGPs manufactured according to embodiments outperform copper. From these results, the thin TGP's, effective thermal conductivity may be between 500 and 5,000 W/mK or 1,000 and 1,500 W/mK.

In some embodiments, the location of heat extraction condensation may affect a TGP's performance. FIG. 13 shows an experimental setup that can be used to characterize the properties of a thin flexible TGP with distributed condensers condensation cooled by air nature air convection without an active cooling heat sink according to some embodiments. In the Figure, a 1 inch $\times$ 1 inch heater placed at the center of each sample's bottom side with wicking layer attached, the temperature difference between an upper region and the center (the heated region) on the top side with spacer layer attached was about 0.6° C. for the TGP sample with 4 W input. This temperature difference was increased to 5.4° C. for the copper sample with the same 4 W input. With distributed condensation over a large area, comparing to copper, the heat spreading performance of the TGP's was improved significantly, even without an active heat sink.

FIG. 14A illustrates a TGP 400 according to some embodiments. TGP 400 may be constructed with any of the following configurations. For example, the material on the interior surface exposed to water may be copper. As another example, a stainless steel mesh 405 may be used that is

encapsulated by copper. As another example, a copper mesh can be used. As another example, the spacer **420** for the vapor core may be defined by a polymer mesh such as, for example, a nylon mesh or a PEEK mesh in conjunction with or in addition to copper pillars. In addition the spacer **420** may be a copper mesh. In some embodiments, one or more copper clad Kapton layers **410** and **420** may be included. In some embodiments, a wicking layer **405** may be included. One or more layers may be combined and/or bonded together.

FIG. **14B** shows an example TGP according to some embodiments. In this example, a plurality of meshes (e.g., a copper mesh and a nylon/peek spacer) may be sealed between copper clad Kapton layers.

The bottom layer and/or the top layer may be assembled and/or sealed with one or more sealing techniques such as, for example, ultrasound welding, laser welding, pressure thermo-compression between two layers and/or with a solder seal to form a high-yield hermetic seal. Any other technique may be used to hermetically seal the TGP such as, for example, photolithography-defined solder masking, copper-to-copper seam welding including ultrasonic welding or laser welding, copper-silver sintering, solder dipping, etc. In this example, a first seal **455** and a second seal **455** may be used.

FIG. **15** illustrates another example TGP according to some embodiments. In this example, a bottom layer of copper clad Kapton and a top layer of copper clad Kapton can be bonded by copper silver sintering using a silver paste. The silver paste, in some embodiments, may allow for lower temperature soldering or sintering. For example, silver paste may be used for bonding at 250° C. without any pressure applied. In some embodiments, the copper surface may be treated with gold, silver and/or palladium. In some embodiments, the organic binder in the paste may be removed by a drying process before sintering.

FIG. **16** illustrates an example process for fabricating a TGP according to some embodiments. At block **505**, a plurality of pillars may be formed on a bottom layer. These pillars, for example, may be used to form a liquid layer or channels. The pillars may be formed using any type of lithographic patterning process or mechanical scribing process. The pillars may have a height less than a millimeter, a diameter of less than 5 millimeters, and a spacing between pillars that is greater than the diameter of the individual pillars. The pillars may be metal and/or polymers based, and/or include a hydrophilic coating or a hydrophobic coating.

At block **510**, a plurality of spacers may be formed on a top layer. These spacers, for example, may be used to form a vapor core layer. The spacers may be formed using any type of lithographic patterning process. The spacers may have a height less than a millimeter, a diameter of less than 5 millimeters, and a spacing between pillars that is greater than the diameter of the individual pillars. In some embodiments, the spacers may have a diameter of less than the diameter of the pillars and/or a space between pillars that is greater than the diameter of the individual pillars. The spacers may be metal and/or polymer based and/or encapsulated by copper and/or include a hydrophilic coating and/or include a hydrophobic coating.

At block **515**, a mesh layer may be sandwiched between the top layer and the bottom layer. The mesh may be a metal mesh, a metal-encapsulated mesh, or a copper-encapsulated stainless steel mesh. The mesh may be a woven mesh. The woven mesh, for example, may have a weave that is less than 75 microns in thickness. In some embodiments, the

mesh may be copper encapsulated. In some embodiments, the mesh may have a desirable wettability with hydrophilic or hydrophobic coatings. In some embodiments, the reaction of the mesh with water may be negligible. In some embodiments, the mesh can be bonded to the bottom layer by electroplated copper.

At block **520** the top layer and the bottom layer may be sealed to create a void cavity within which a working fluid may be placed. Any type of sealing may be used including those described herein and those not described herein. A tube may extend through the sealing to allow evacuation and charging of the TGP.

At block **525** any air or non-condensable gas or other material within the void may be evacuated using any technique. At block **530** the TGP may be charged with a working fluid such as, for example, water, methane, ammonia, or other working fluid that is compatible with the exposed TGP inner surface.

The various steps in the process shown in FIG. **16** may occur in any order and/or any block may be removed.

In some embodiments, a TGP may be sealed with tin/lead solder (or similar lead-free solder alloys) according to some embodiments. Tin/lead solder's reaction with water may be limited.

FIG. **17** illustrates a top view of another TGP **700** and FIGS. **18A**, **18B**, and **18C** illustrate side views of the TGP **700** according to some embodiments. The TGP **700** includes a top layer (**805** in FIG. **18**) and a bottom layer **705**. FIG. **17** shows the TGP **700** with the top layer **805** removed. The top layer **805** and/or the bottom layer **705** may include copper and/or polyimide material. In some embodiments, the top layer **805** and/or the bottom layer **705** may include layers of both copper and polyimide.

The TGP **700** includes a mesh return layer **715** that includes a plurality of return arteries **710** formed or cut within the mesh return layer **715**. In some embodiments, the return arteries **710** may not extend into the evaporator region **720** of the TGP **700**. In some embodiments, the width of the return arteries **710** may be less than 30 microns. In some embodiments, the width of the return arteries **710** may be less than 100 microns. In some embodiments, the mesh return layer **715** may include a wick. In some embodiments, the mesh return layer **715** may include a steel mesh such as, for example, a mesh with wires that are less than 50 microns or 25 microns thick. In some embodiments, the mesh return layer **715** may include a mesh with a simple weave at **200**, **300**, **400**, **500**, **600**, **700**, etc. wires/inch. In some embodiments, the mesh return layer **715** may be electroplated with copper. The mesh return layer **715** may include a specified evaporator region that may provide a specific location for a heat source. In some embodiments, the mesh return layer **715** may have a thickness of about 40 microns.

The mesh return layer **715** may have any number of shapes and/or configurations. In some embodiments, the mesh return layer **715** may have a polygonal or circular shape. In some embodiments, the mesh return layer **715** may have multiple sections without return arteries **710**. In some embodiments, the mesh return layer **715** may include any number, shape or configuration of return arteries **710**. In some embodiments, the return arteries **710** may have one or more pillars or other mechanisms disposed within the return arteries **710**.

In some embodiments, return pillars **835** (see FIG. **18**) may extend from the bottom layer **705** through the return arteries **710** of the mesh return layer **715**. In some embodiments, the return pillars **835** may form vapor regions running parallel to the mesh return layer **715**. In some embodi-

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ments, the return pillars **835** may form arteries in the adiabatic and/or condenser regions of the TGP **700**.

The return pillars **835** may have at least one dimension that is smaller than the width of the return arteries **710**. In some embodiments, these pillars may have at least one dimension (e.g., height, width, length, diameter, etc.) that is less than 10 microns. In some embodiments, these pillars may have at least one dimension (e.g., height, width, length, diameter, etc.) that is less than 50 microns. In some embodiments, these pillars may have at least one dimension (e.g., height, width, length, diameter, etc.) that is less than 100 microns.

FIG. **18A** shows a side view of TGP **700** cut through Section A shown in FIG. **17**. In FIG. **18A**, the TGP **700** is cut through a region where a return arteries **710** extends along a portion of the mesh return layer **715**. As shown in FIG. **18A**, the mesh return layer **715** is present in the evaporator region. The return pillars **835** are shown extending through the return arteries **710**. The TGP **700** also includes a plurality of top pillars **825** disposed on the top layer **805**. The top pillars **825** may have at least one dimension (e.g., height, width, length, diameter, etc.) that is larger than the return pillars **835**. The top pillars **825** may have at least one dimension (e.g., height, width, length, diameter, etc.) that is larger than 0.25 mm, 0.5 mm, 0.75 mm, 1.0 mm, 1.25 mm, etc.

In some embodiments, the TGP **700** may include a micro wick layer **815**. The micro wick layer **815**, for example, may include a plurality of pillars (e.g., electroplated pillars). The micro wick layer **815** may have at least one dimension (e.g., pillar height, width, length, diameter, pitch, etc.) that is smaller than the return pillars **835**. The micro wick layer **815** may have at least one dimension (e.g., height, width, length, diameter, etc.) that is smaller than 5 μm , 10 μm , 15 μm , 20 μm , 25 μm , etc. The micro wick layer **815** may be aligned with the return arteries **710**.

FIG. **18B** shows a side view of TGP **700** cut through Section B shown in FIG. **17**. In FIG. **18B**, the TGP is cut through a region without the return arteries **710** extending along a portion of the mesh return layer **715**. Instead, the mesh return layer **715** extends along the length of the TGP **700** along this section of the TGP **700**.

FIG. **18C** shows an end view of TGP **700** cut through Section C shown in FIG. **17**. In FIG. **18C**, the TGP is cut through the mesh return layer **715** showing both the mesh return layer **715** and the return arteries **710** formed in the mesh return layer **715**. In some embodiments, the return pillars **835** may extend through the return arteries **710**. In some embodiments, one or more of the return pillars **835** may contact one or more of the top pillars **825**.

In some embodiments, the top layer **805** and the bottom layer **705** are sealed along at least one edge of the top layer **805** and along at least one edge of the bottom layer **705**. In some embodiments, the top layer **805** and the bottom layer **705** are sealed along at least two edges of the top layer **805** and along at least two edges of the bottom layer **705**.

In some embodiments, as shown in FIG. **19**, the plurality of wicking structures **990** comprise a mesh layer. The mesh layer, for example, may include a mesh selected from the list consisting of copper mesh, stainless steel mesh, metal mesh, polymer mesh and copper-encapsulated mesh. The mesh layer, for example, may include a hydrophilic coating **995** or a hydrophobic coating or hermetic coating.

FIG. **20** illustrates a TGP **2000** according to some embodiments. In this example, the top layer **310** and/or the bottom layer **315** may include copper clad Kapton and/or may be similar to the top layer **310** and the bottom layer **315** described above in conjunction with FIG. **3**. In some

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embodiments, the top layer **310** and the bottom layer **315** of the TGP **2000** may be sealed together using solder, laser welding, ultrasonic welding, electrostatic welding, or thermo-pressure compression. In between the top layer **310** and the bottom layer **315** the TGP **600** may include a liquid channel, a wicking structure, and/or a vapor core.

In some embodiments, the top layer **310** and/or the bottom layer **315** may comprise copper and stainless steel, copper and a polymer, copper, copper clad Kapton, polymer, etc. In some embodiments, the top layer **310** and the bottom layer **315** may include the same material or different material.

The mesh **610**, for example, may include a copper mesh, stainless-steel mesh with or without a copper coating, or one or more mesh made of other materials but encapsulated with copper. The, for example, may include one, two, three, four, five, six or more layers of mesh of the same or different material. In some embodiments, the may be bonded and/or sealed using FEP (fluorinated ethylene propylene) electroplating, sintering, and/or other adhesives or sealants with the top layer **310**. In some embodiments, in order to enhance hydrophilic properties of the, atomic layer deposition (ALD) of Al_2O_3 , TiO_2 , SiO_2 , or other coatings may be used to encapsulate the plurality of pillars of the liquid channel **615** with desirable functionality. In some embodiments, the may be bonded with or on a plurality of pillars of the liquid channel **615**.

In some embodiments, the mesh **610** may include a woven mesh that may be bonded to the second plurality of pillars of the liquid channel **615** and/or the vapor core **605** through an electroplating process. In some embodiments, the mesh may be a copper woven mesh or a metallic mesh. During this process, for example, the (e.g., woven mesh) may be encapsulated by copper. The top layer **310** with the vapor core **605** and the bottom layer **315** with the liquid channel **615** may be sealed together with the with a mesh in between. After sealing, the TGP **600** may be charged with a working fluid, such as water, methane, ammonia, or other coolants or refrigerants that are compatible with the surfaces exposed to the working fluid.

The liquid channel **615**, for example, may include a first plurality of pillars (e.g., electroplated pillars).

The vapor core **605**, for example, may include a second plurality of pillars (e.g., electroplated pillars). In some embodiments, the second plurality of pillars in the vapor core **605** may have at least one dimension (e.g., height, width, length, diameter, etc.) that is smaller than at least one dimension (e.g., height, width, length, diameter, etc.) of the second plurality of pillars of the liquid channel **615**. The second plurality of pillars may be disposed on the bottom layer **315**. The second plurality of pillars may extend to the mesh **610** and/or may not extend to the top layer **310**.

In some embodiments, a third plurality of pillars **620** may extend from the bottom layer **315** to the top layer **310**. The third plurality of pillars **620** may extend through or around the mesh **620**.

In some embodiments, a photo-lithographic patterning process may be used to fabricate the pillars in the liquid channel **615** and/or the vapor core **605**. For example, a photo-lithography process may be used to form a plurality of pillars on the top layer **310** to form the vapor core **605** and/or a photo-lithographic patterning process may be used to form a plurality of pillars on the bottom layer **315** to form the liquid channel **615**. The photo-lithographic patterning process, for example, may control the positioning and/or the heights of the pillars with a resolution of a few microns. The

plurality of pillars on the top layer **310** and/or on the bottom layer **315** may be formed by the mechanical scribing process.

In some embodiments, the thickness of a TGP (e.g., TGP **300** or TGP **600**, or any other TGP described in this disclosure) may be a few millimeters or even less than about 500, 450, 400, 350, 300, 250, 200, 150, or 100 microns. For example, the top layer **310** and/or the liquid channel **615** may be 5, 10, 20, 30, 40, 50, 60, etc. microns thick, the wicking structure (e.g.,) may be 5, 10, 20, 30, 40, 50, 60, etc. microns thick, and/or the vapor core (e.g., vapor core **605**) may be 10, 20, 30, 40, 50, 75, 100, 125, 150, etc. microns thick.

In some embodiments, a buffer region can be created by design to collect and store any non-condensable gas through passive convection. For example, a space of a few millimeters can be formed in the area outside the mesh (e.g., outside the mesh region shown in FIG. **6**). This space can be added prior to bonding. This space may collect any non-condensable gases that would move to this space because of its different density, and thus its effect on evaporation and condensation can be reduced substantially.

Various other sealing techniques may be used such as, for example, thermosonic or thermo-compression bonding, ultrasonic welding, laser welding, electron beam welding, electroplating; solder sealing with alloys with negligible reaction with water; and polymer bonding encapsulated by moisture barrier coatings such as atomic layer deposition (ALD)-based coatings.

Some embodiments may include a pillar-enabled TGP. In some embodiments, the TGP may include a copper-cladded Kapton film that includes three layers. These layers may, for example, include copper and Kapton layers. Each layer may be about 12 μm thick. In some embodiments, a stainless steel woven mesh may be included and may have a thickness less than 75 μm . In some embodiments, the pillars may allow for fluid and/or vapor transport between the pillars under different mechanical loadings.

In some embodiments, a plurality of pillars may be formed on a copper layer (e.g., the top layer and/or the bottom layer) using any of various lithography lithographic patterning processes.

In some embodiments, a copper-encapsulated stainless steel mesh may be sandwiched between the top layer and the bottom layer. The stainless steel mesh, for example, may have a weave that is less than 75 microns in thickness. In some embodiments, the mesh may be copper encapsulated. In some embodiments, the mesh may be hydrophilic. In some embodiments, the reaction of the mesh with water may be negligible.

In some embodiments, a TGP may include a mesh-pillar wicking structure. The mesh-pillar wicking structure may allow the TGP to achieve a low capillary radius (high pumping pressure) in the evaporation regions and/or a higher flow hydraulic radius (low flow pressure drop) in the fluid channel.

In some embodiments, a TGP may include pillars with rounded heads. For example, the pillars may be formed with controlled over-plating. In some embodiments, the pillars may form very sharp angle at the interface between a pillar and the mesh bonded. In some embodiments, these sharp angles may be used, for example, to enhance the capillary pumping force pulling the liquid returned from the condenser to the evaporator.

In some embodiments, a plurality of star-shaped pillars may be constructed on either the top layer and/or the bottom layer that have a star-shaped polygon various cross section.

In some embodiments, a plurality of hydrophilic pillars may be constructed on either the top layer and/or the bottom layer.

In some embodiments, heat rejection through condensation can be distributed throughout the external surface of the TGP.

In some embodiments, pillars and/or spacers may be disposed on a layer with densities (spacing between pillars or spacers) that vary across the layer, with diameters that vary across the layer, with spacing that vary across the layer, etc.

The Figures are not drawn to scale.

The term “substantially” means within 5% or 10% of the value referred to or within manufacturing tolerances.

Numerous specific details are set forth herein to provide a thorough understanding of the claimed subject matter. However, those skilled in the art will understand that the claimed subject matter may be practiced without these specific details. In other instances, methods, apparatuses, or systems that would be known by one of ordinary skill have not been described in detail so as not to obscure claimed subject matter.

The use of “adapted to” or “configured to” herein is meant as open and inclusive language that does not foreclose devices adapted to or configured to perform additional tasks or steps. Additionally, the use of “based on” is meant to be open and inclusive, in that a process, step, calculation, or other action “based on” one or more recited conditions or values may, in practice, be based on additional conditions or values beyond those recited. Headings, lists, and numbering included herein are for ease of explanation only and are not meant to be limiting.

While the present subject matter has been described in detail with respect to specific embodiments thereof, it will be appreciated that those skilled in the art, upon attaining an understanding of the foregoing, may readily produce alterations to, variations of, and equivalents to such embodiments. Accordingly, it should be understood that the present disclosure has been presented for purposes of example rather than limitation, and does not preclude inclusion of such modifications, variations, and/or additions to the present subject matter as would be readily apparent to one of ordinary skill in the art.

That which is claimed:

1. A thermal ground plane comprising:

a first substrate member comprising copper;

a second substrate member comprising a metal, wherein the first substrate member and the second substrate member are sealed together so as to be in direct contact and to enclose a working fluid;

a first plurality of pillars disposed on an interior surface of the first substrate;

a mesh layer disposed on the top of the first plurality of pillars, wherein the mesh layer comprises at least one of copper, polymer encapsulated with copper, or stainless steel encapsulated with copper, wherein the mesh is entirely disposed in a cavity defined by the first substrate member and the second substrate member; and

a second plurality of pillars disposed on an interior surface of the second substrate member within an area defined by the perimeter of the second substrate member and the second plurality of pillars extend from the second substrate member to the mesh layer.

2. The thermal ground plane according to claim 1, wherein the second plurality of pillars are not in contact with the first substrate member.

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3. The thermal ground plane according to claim 1, wherein the thermal ground plane has a thickness less than 300 microns.

4. The thermal ground plane according to claim 1, wherein the mesh layer is bonded with the first plurality of pillars.

5. The thermal ground plane according to claim 1, wherein the mesh layer comprises at least two mesh layers and each mesh layer comprises the same material as the other mesh layers.

6. The thermal ground plane according to claim 1, wherein the mesh layer comprises at least two mesh layers and each mesh layer comprises a different material than the other mesh layers.

7. The thermal ground plane according to claim 1, wherein a perimeter of the first substrate member and a perimeter of the second substrate member are hermetically sealed together.

8. The thermal ground plane according to claim 7, wherein the hermetical seal is a thermocompression-bonded, an ultrasound-welded, an electrostatic welded, or a laser-welded copper-to-copper interface.

9. The thermal ground plane according to claim 1, wherein the spacing between each pillar of the second plurality of pillars is larger than two times of the diameter of each pillar of the second plurality of pillars.

10. The thermal ground plane according to claim 1, wherein the mesh layer may include a mesh selected from the list consisting of copper mesh, stainless steel mesh, metal mesh, polymer mesh and copper-encapsulated mesh.

11. A thermal ground plane comprising:

a first substrate member comprising at copper and either stainless steel or polymer;

a second substrate member comprising a metal, wherein a perimeter of the first substrate member and a perimeter of the second substrate member are hermetically sealed together so as to be in direct contact, wherein the first substrate member and the second substrate member enclose a working fluid;

a mesh layer bonded with the first substrate member, wherein the mesh layer comprises at least one of copper, polymer encapsulated with copper, or stainless steel encapsulated with copper, wherein the mesh is entirely disposed in a cavity defined by the first substrate member and the second substrate member; and

a plurality of pillars disposed on an interior surface of the second substrate member within an area defined by the perimeter of the second substrate member and the second plurality of pillars extend from the second substrate member to the mesh layer and are not in contact with the first substrate member.

12. The thermal ground plane according to claim 11, wherein the mesh layer comprises at least two mesh layers and each mesh layer comprises the same material as the other mesh layers.

13. The thermal ground plane according to claim 11, wherein the mesh layer comprises at least two mesh layers and each mesh layer comprises a different material than the other mesh layers.

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14. The thermal ground plane according to claim 11, wherein the thermal ground plane has a thickness less than 300 microns.

15. The thermal ground plane according to claim 11, wherein the mesh layer may include a mesh selected from the list consisting of copper mesh, stainless steel mesh, metal mesh, polymer mesh and copper-encapsulated mesh.

16. A thermal ground plane comprising:

a first substrate member comprising copper;

a second substrate member comprising a metal, wherein the first substrate member and the second substrate member are sealed together so as to be in direct contact and to enclose a working fluid;

a mesh layer bonded with the first substrate member, wherein the mesh layer comprises at least one of copper, polymer encapsulated with copper, or stainless steel encapsulated with copper, wherein the mesh is entirely disposed in a cavity defined by the first substrate member and the second substrate member; and

a second plurality of pillars disposed on an interior surface of the second substrate member within an area defined by the perimeter of the second substrate member and the second plurality of pillars extend from the second substrate member to the mesh layer and are not in contact with the first substrate member.

17. The thermal ground plane according to claim 16, wherein the mesh layer comprises at least two mesh layers and each mesh layer comprises the same material as the other mesh layers.

18. The thermal ground plane according to claim 16, wherein the mesh layer comprises at least two mesh layers and each mesh layer comprises a different material than the other mesh layers.

19. A thermal ground plane comprising:

a first substrate member comprising copper;

a second substrate member comprising a metal, wherein the first substrate member and the second substrate member are sealed together so as to be in direct contact and to enclose a working fluid;

a mesh layer bonded with the first substrate member, wherein the mesh layer comprises at least one of copper, polymer encapsulated with copper, or stainless steel encapsulated with copper, wherein the mesh is entirely disposed in a cavity defined by the first substrate member and the second substrate member; and

a second plurality of pillars disposed on an interior surface of the second substrate member within an area defined by the perimeter of the second substrate member and the second plurality of pillars extend from the second substrate member to the mesh layer.

20. The thermal ground plane according to claim 19, wherein the mesh layer comprises at least two mesh layers and each mesh layer comprises the same material as the other mesh layers.

21. The thermal ground plane according to claim 19, wherein the mesh layer comprises at least two mesh layers and each mesh layer comprises a different material than the other mesh layers.

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